

Comp 295

Intro to C++ - Spring 2023

Project Document

Instructions

1. Work on this Project in group of two students.
2. Any traces of plagiarism would result in a ZERO marks and possible disciplinary action.
3. Submit the UML diagram by 28th May.
4. Project submission deadline is Friday 9th June.

Group details:

<https://docs.google.com/spreadsheets/d/1T9G430o5mNF6NuyzG2bvjTysUI0MZmTeWqa-ORTfeeU/edit?usp=sharing>

Goal

The aim of the project is to learn and create a practical application using a programming approach called object-oriented programming in the C++ language. The project idea is similar to what you previously worked on in the Programming 2 project. You can use the same plan and write the code in C++ for this project.

In the project, you will be evaluated based on your:

1. Understanding of the OOP concepts.
2. Implementation of the algorithms and OOP concepts.
3. Ability to observe real-world things in terms of objects and classes.
4. Encapsulation: Your ability to encapsulate data and methods within classes, ensuring that data is hidden and accessed only through defined methods.
5. Inheritance: Demonstrating the use of inheritance to create hierarchical relationships between classes, allowing for code reuse and specialization.
6. Polymorphism: Implementing polymorphic behavior by utilizing function overriding or interfaces to allow objects of different classes to be used interchangeably.
7. Abstraction: Utilizing abstraction to create simplified and generalized class interfaces, hiding complex implementation details, and providing a clear separation between interface and implementation.
8. Modularity: Designing your project with modular and reusable components, allowing for easier maintenance and extensibility. (implement ~ 6-8 classes)
9. Use of Operator Overloading.

Bonus points for GUI implementation

Submission Detail:

- Phase 1 – UML class diagram and Documentation – **Due 28th May,2023**

In the first phase of the project, you are supposed to select the management system. You will clearly identify various classes, their attributes, behaviors and interactions between classes. You will represent all this information in a class diagram and submit the diagram. Keep in mind that program design is fixed once class diagram has been submitted. You can refer to the material provided at the following link while creating your class diagram.

<https://www.tutorialspoint.com/uml/index.htm>

You need to provide a documentation which contains all the details just like it is given below (See sample project-documentation section)

- Phase 2 - Implementation - **Due date 9th June,2023**

You will work on implementing the project in light of your class diagram and will submit the code. Code should be well formatted. It should have sufficient comments. You have to describe the menu of your management system and describe the functionality of each command.

- Phase 3 – Project Viva - **TBA**

Project Viva will be conducted in the classroom. Viva will be conducted on the basis of goals mentioned in above section. Phase 1 and 2 will be marked on the basis of Project Viva.

General Requirements

- Your project should have at least 7 classes.
- Your project should have all the relationships implementation (Association, aggregation or Composition, inheritance with polymorphic Behavior)
- Your project should have a proper Menu
- Data should be loaded into system through filing.
- Every requirement will be evaluated and marked.

Sample project-documentation:

You are required to implement the simple and basic Computer management system (CMS). It should maintain records of: Students, Teachers and Classes

Student: Attributes are Name, ID and CGPA

Teacher: Attributes are Name, ID and Scale (integer type)

Class: Attributes are Students (Strength should be variable but use of memory should be optimized.), ID and Teacher

Write getters, setters, constructors and destructors for each.

Menu

Program should display follow menu:

Press

1: Create new class

2: Swap Teachers

3: Display all CRs

4: Add 2 classes

5: Add Student to a class

6: Delete Student

7: Display Class

8: Show the better class

9: Show Student information

10: Show Teacher information

Here is the explanation to each.

Working of Menu:

Create Class:

When user enters 1, He/she should be asked for class id, class teacher Information, no. of students and their Information.

After giving all this information, class should be created.

Swap Teachers:

When user enters 2, program should display all class ids and ask user to enter two of them so that the teachers of both classes may be swapped with each other.

Show All CRs:

When user enters 3, program should display all CRs.

Add 2 Classes:

When user enters 4, program should display all class ids and ask user to enter 2 of those ids and then add these two classes to generate a new class. The teacher for this new class will be the one with greater scale. And Students with highest CGPAs will be added in this class.

Add Student to a Class:

When user enters 5, program displays all available classes. Then it should ask user to enter the class id of that class in which you want to add new student. Handle all possible wrong entries.

Delete Student:

When user enters 6, program should ask user to enter student id whom you want to delete. When user enters the id then program first displays all information of student and then delete this.

Display Class:

When user enters 7, program should ask class id, and then display all information of the class whose id is entered.

Show the Best Class:

When user enters 8, program should display all available classes and then ask user for 2 classes and then program will display the best class among these. The criteria for better class are, class with more no. of students having CGPA higher than 3.00 is better among these. Program should handle unusual case if there is.

Show Student Information:

When user enters 9, program should ask for Student id, after that it should display all information of student whose id is entered (name, CGPA, his class id and his teacher name).

Show Teacher Information:

When user enters 10, program should ask for Teacher id, after that it should display all information of Teacher whose id is entered (name, class, his class id and student names in his class).

Note:

All class data members should not be access directly.